Written Work

3) At $\omega \to \infty, Z_C \cong 0 \to V^+ = V^- = 0$

$\omega \to 0, Z_C \cong \infty \to V^+ = V^- = 0$ (pull-down)

BPF

$$\frac{V_x - V_o}{(A-1)R} + \frac{V_x}{R} = 0 \to V_x\left(\frac{1}{(A-1)R} + \frac{1}{R}\right) = \frac{V_o}{(A-1)R}$$

$$V_x = V_o\left(\frac{1}{(A-1)} \div \left[\frac{1}{(A-1)R} + \frac{1}{R}\right]\right) = \frac{V_o}{1+A-1} = \frac{V_o}{A}$$

$$\frac{V_x - V_y}{1/sC} + \frac{V_x}{2R} \to V_x\left(sC + \frac{1}{2R}\right) = V_y sC \to \frac{V_o\left(sC + \frac{1}{2R}\right)}{AsC} = V_y$$

$$\frac{V_o}{A}\left(\frac{1}{2sRC} + 1\right) = V_y$$

$$(V_y - V_x)sC + V_y sC + \frac{V_y - V_i}{R} + \frac{V_y - V_o}{R} = 0$$

$$V_y\left(2sC + \frac{2}{R}\right) - V_x sC - V_i\left(\frac{1}{R}\right) - V_o\left(\frac{1}{R}\right) = 0$$

$$-V_o/A \, sC$$

$$V_y\left(2sC + \frac{2}{R}\right) - V_i\left(\frac{1}{R}\right) = V_o\left(\frac{sC}{A} + \frac{1}{R}\right) = 0$$

$$\frac{V_o}{A}\left(\frac{1}{2sRC} + 1\right)\left(2sC + \frac{2}{R}\right) - V_o\left(\frac{sC}{A} + \frac{1}{R}\right) = \frac{V_i}{R}$$

$$\frac{V_o}{A}\left(\frac{1}{R} + \frac{1}{sR^2C} + 2sC + \frac{2}{R} - sC - \frac{1}{R}\right) = V_i/R$$

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$$\frac{V_o}{A}\left(\frac{1}{sR^2C}+2sC+\frac{3}{R}\right)-V_o\left(\frac{sC}{A}+\frac{1}{R}\right)=\frac{V_i}{R}$$

$$\frac{V_o}{AsR^2C}+\frac{2V_osC}{A}+\frac{3V_o}{AR}-\frac{V_osC}{A}-\frac{V_o}{R}=\frac{V_i}{R}$$

$$\frac{V_o}{AsRC}+\cancel{\frac{2V_osRC}{A}}+\frac{3V_o}{A}\cancel{-\frac{V_osRC}{A}}-V_o=V_i$$

$$V_o\left(\frac{1}{AsRC}+\frac{sRC}{A}+\frac{3}{A}-1\right)=V_i$$

$$H(s)=\frac{1}{\frac{1}{AsRC}+\frac{sRC}{A}+\frac{3}{A}-1}\rightarrow\cdot\frac{sA}{RC}\Big/\frac{sA}{RC}=\frac{sA/RC}{s^2+s\left(\frac{3}{RC}-\frac{A}{RC}\right)+\frac{1}{(RC)^2}}$$

$Ks/Q\omega_o$

$$=\frac{\frac{sA}{RC}}{s^2+s\left(\frac{3-A}{RC}\right)+\frac{1}{(RC)^2}}$$

$s^2+s/Q\omega_o+\omega_o^2$

$\frac{1}{QRC}$

$\omega_o=\frac{1}{RC}$

$\rightarrow Q=\frac{1}{3-A}$

$K=\frac{A}{3-A}$

$$\frac{k}{Q}=A\rightarrow k=AQ=A\left(\frac{1}{3-A}\right)=\boxed{\frac{A}{3-A}}$$

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4) $\frac{V_o - V_x}{R_2} + V_o s C_2 = 0 \rightarrow V_o\left(sC_2 + \frac{1}{R_2}\right) = \frac{V_x}{R_2} \rightarrow V_x = V_o(sR_2C_2 + 1)$

$(V_x - V_o)sC_1 + \frac{V_x - V_o}{-R_2} + \frac{V_x - V_i}{R_1} = 0$

$V_x\left(sC_1 + \frac{1}{R_2} + \frac{1}{R_1}\right) - V_o\left(sC_1 + \frac{1}{R_2}\right) = \frac{V_i}{R_1}$

$V_x\left(sR_1C_1 + \frac{R_1}{R_2} + 1\right) - V_o\left(sR_1C_1 + \frac{R_1}{R_2}\right) = V_i$

$\left(V_o(sR_2C_2 + 1)\left(sR_1C_1 + \frac{R_1}{R_2} + 1\right)\right) - V_o\left(sR_1C_1 + \frac{R_1}{R_2}\right) = V_i$

$\left(V_o\left(s^2R_1R_2C_1C_2 + sR_1C_2 + sR_2C_2 + sR_1C_1 + \frac{R_1}{R_2} + 1 - sR_1C_1 - \frac{R_1}{R_2}\right)\right) = V_i$

$V_o\left(s^2R_1R_2C_1C_2 + sR_1C_2 + sR_2C_2 + 1\right) = V_i$

$H(s) = \frac{1}{s^2R_1R_2C_1C_2 + sR_1C_2 + sR_2C_2 + 1} = \frac{\frac{1}{R_1R_2C_1C_2}}{s^2 + s\left(\frac{1}{R_1C_1} + \frac{1}{R_2C_1}\right) + \frac{1}{R_1R_2C_1C_2}}$ LEFT CKT

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$$V_o s C_2 = -V_x / R_2 \rightarrow V_x = -V_o s R_2 C_2$$

$$\frac{V_x - V_i}{R_1} + \frac{V_x}{R_2} + \frac{V_x - V_o}{R_3} + V_x s C_1$$

$$V_x\left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + sC_1\right) - V_o\left(\frac{1}{R_3}\right) = \frac{V_i}{R_1}$$

$$\Rightarrow V_x\left(\frac{R_1}{R_2} + \frac{R_1}{R_3} + sR_1C_1 + 1\right) - V_o\left(\frac{R_1}{R_3}\right) = V_i$$

$$\Rightarrow -V_o(sR_2C_2)\left(\frac{R_1}{R_2} + \frac{R_1}{R_3} + sR_1C_1 + 1\right) - V_o\left(\frac{R_1}{R_3}\right) = V_i$$

$$\Rightarrow -V_o\left(sR_1C_2 + s\frac{R_1R_2C_2}{R_3} + s^2R_1R_2C_1C_2 + sR_2C_2 + \frac{R_1}{R_3}\right) = V_i$$

$$\Rightarrow H(s) = -\frac{1}{s^2R_1R_2C_1C_2 + s\left(\frac{R_1R_2C_2}{R_3} + R_1C_2 + R_2C_2\right) + \frac{R_1}{R_3}}$$

$$\Rightarrow H(s) = -\frac{1/R_1R_2C_1C_2}{s^2 + s\left(\frac{1}{R_3C_1} + \frac{1}{R_2C_1} + \frac{1}{R_1C_1}\right) + \frac{1}{R_2R_3C_1C_2}}$$ MFB

FIVE STAR

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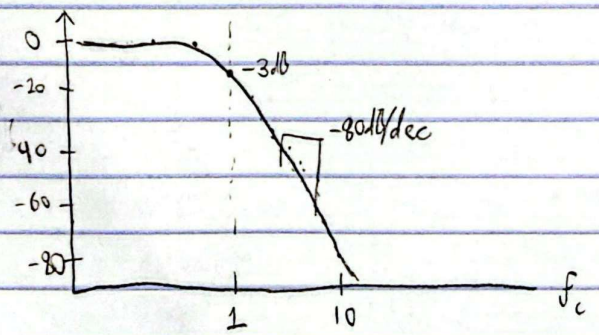
google formula ↓

5) a: $20\log_{10}(2^{12}) \approx 72.25\,dB$

b: No, the V_i and V_o both go thru the LPF, so both have equivalent phase distortion; cancels out in V_o/V_i

c: |H| (dB) 4th order

4 · -20dB/dec = -80dB/dec



d: We want our signal far enough past f_c that f_s is not at attenuation smaller than -72.25dB, so about 72dB = 3 octaves (3·24dB):

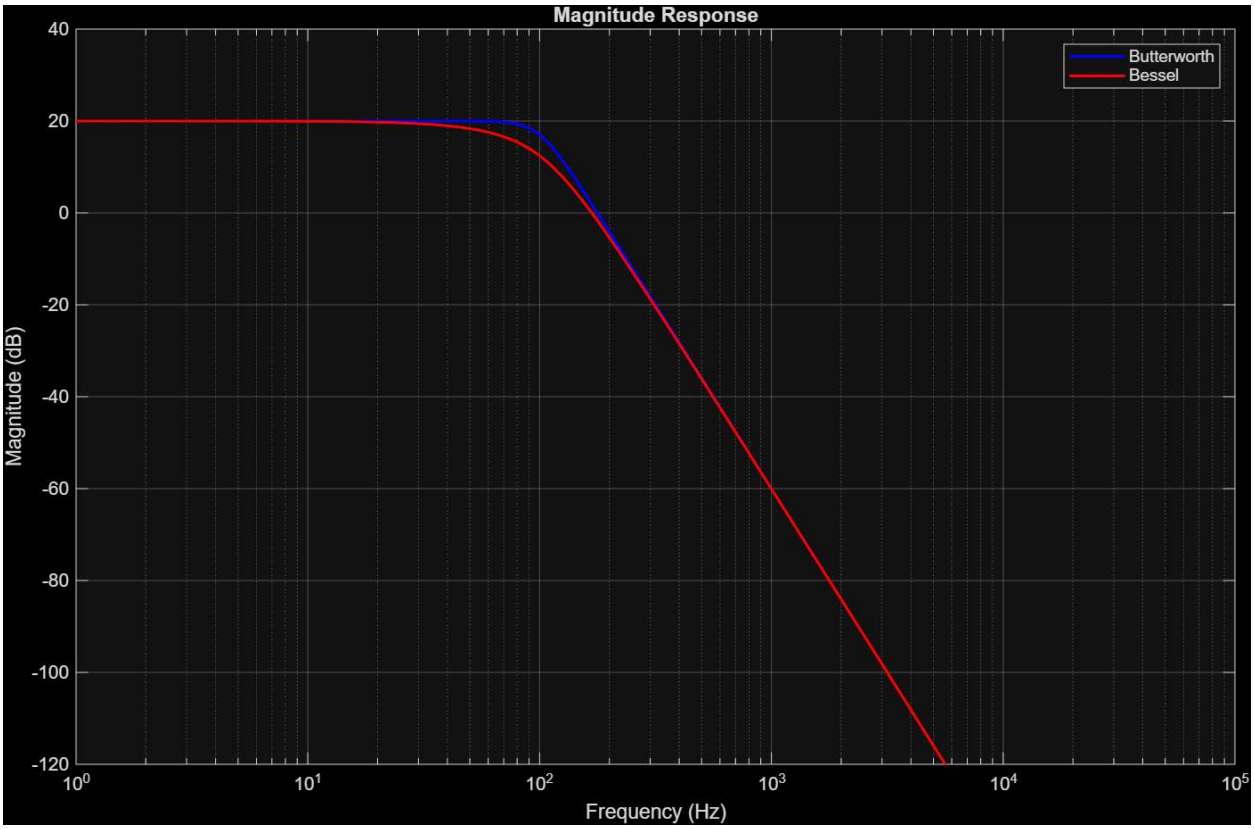
$2000 \cdot 2^3 = 2000 \cdot 8 \approx 16kHz$

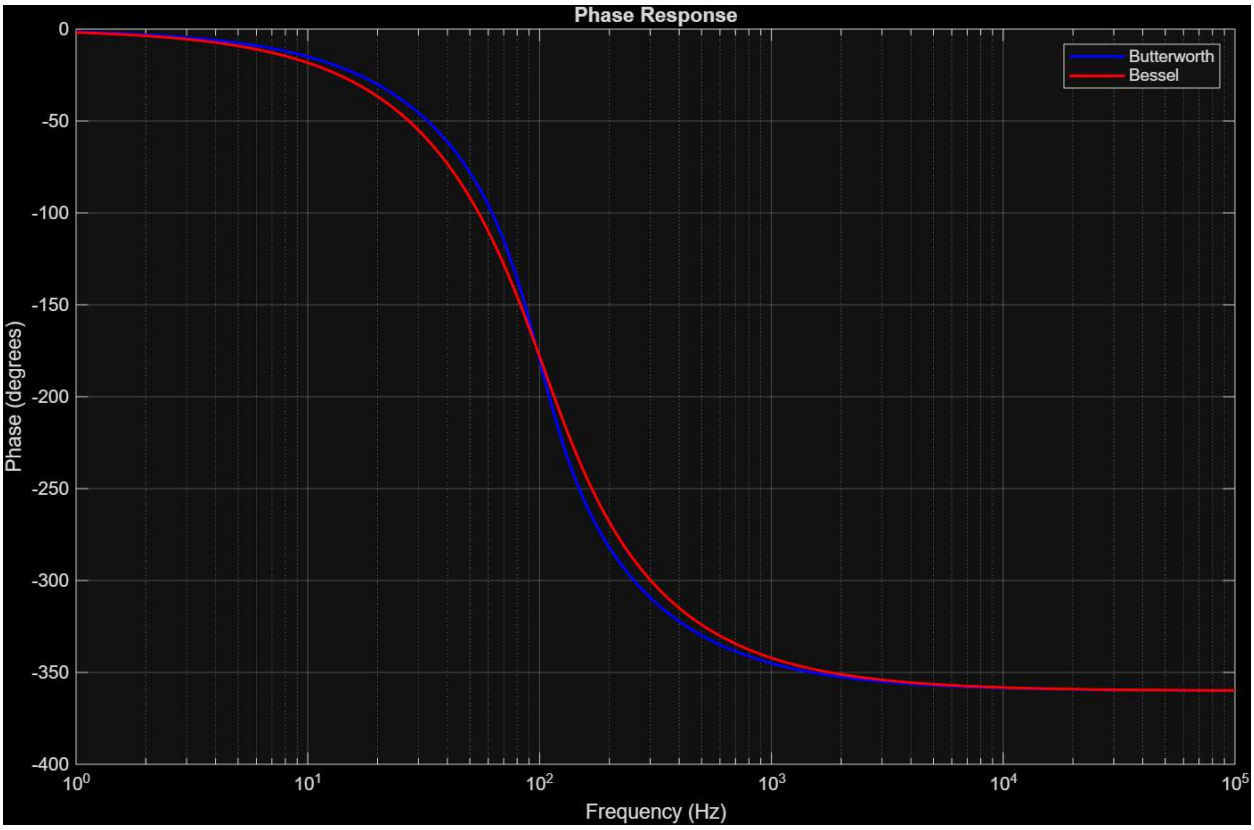
$f_s - 16k \geq 2k$

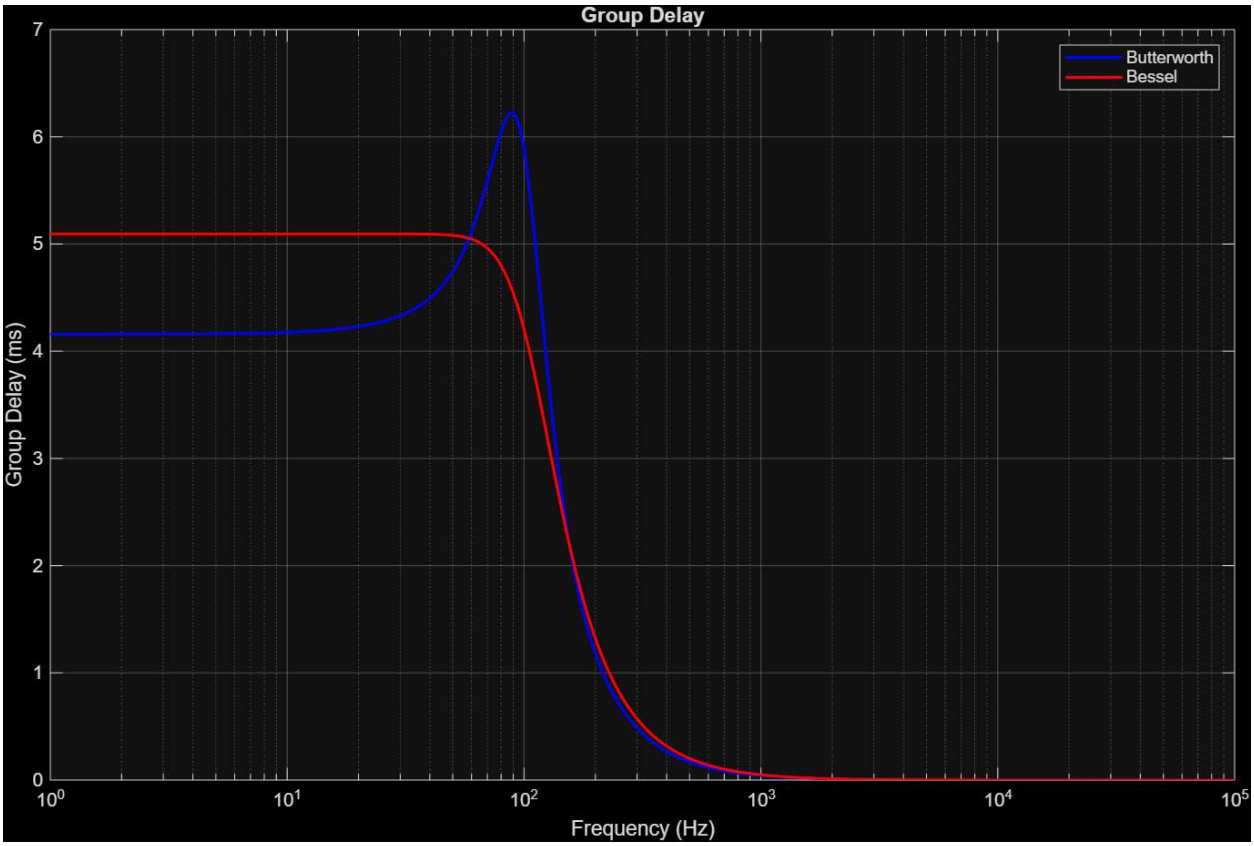
$18000 = f_s$

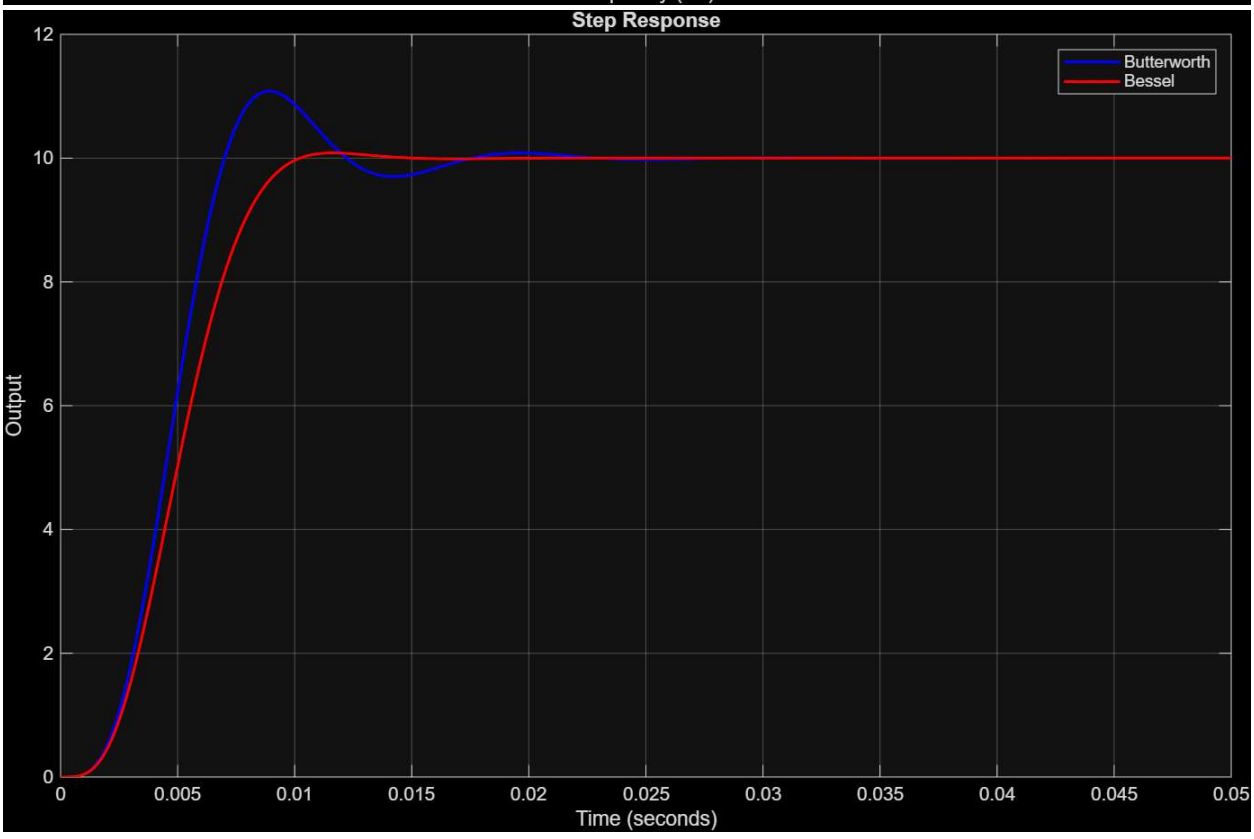
Scanned with CamScanner

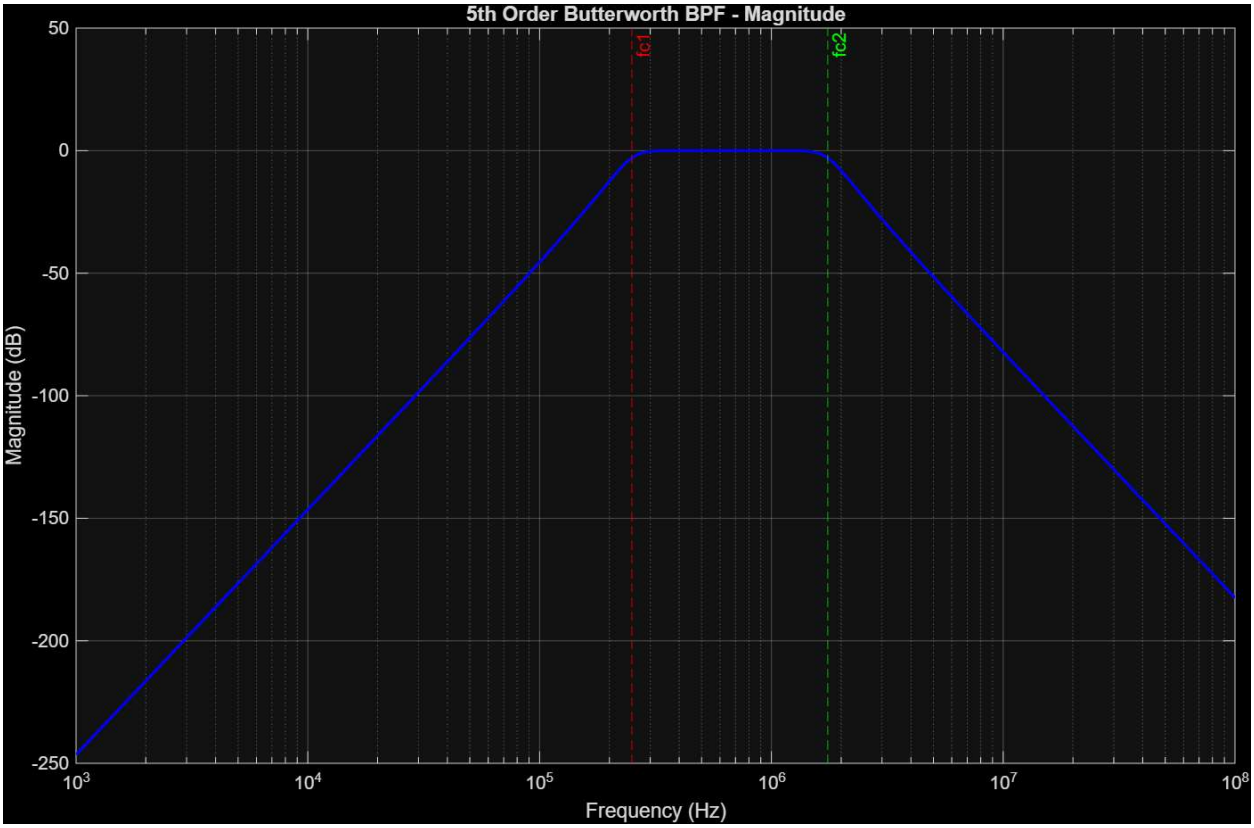
MATLAB Figures

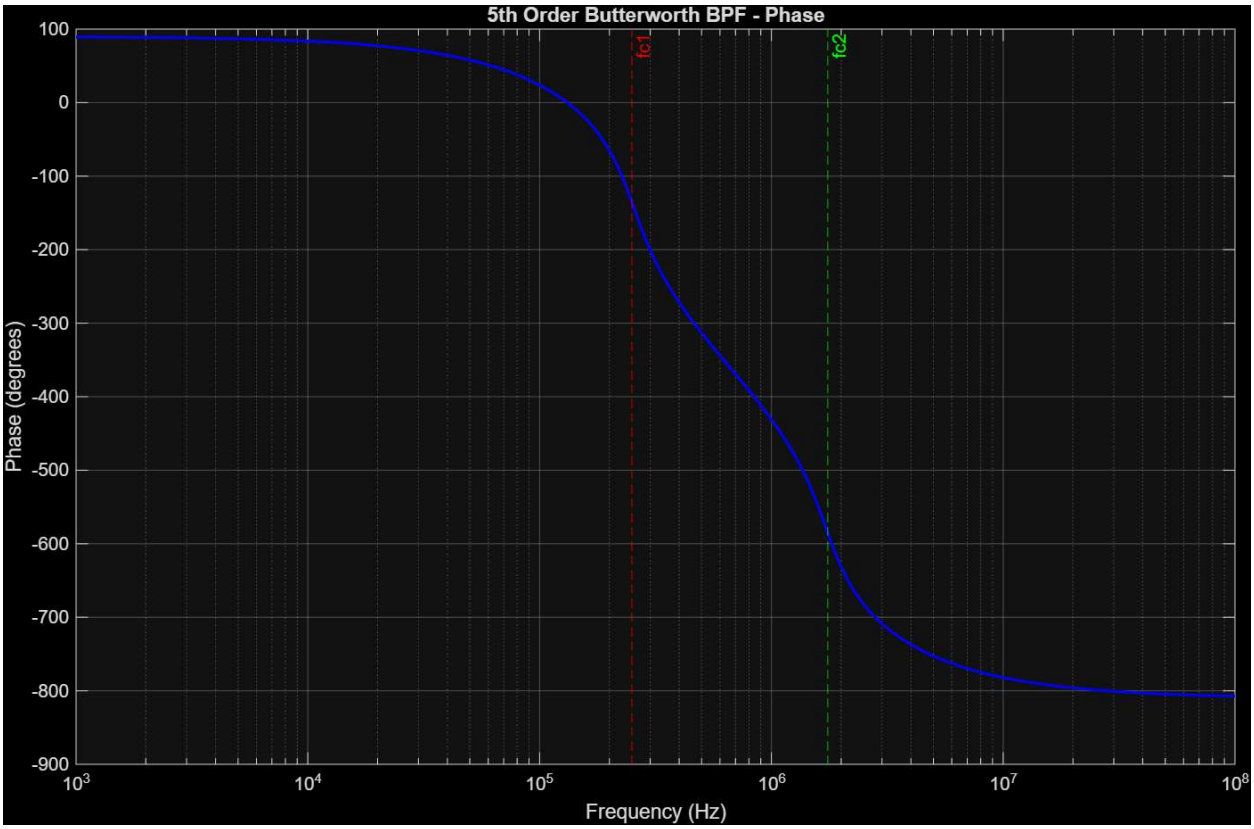





Group Delay
Butterworth
Bessel
Group Delay (ms)
Frequency (Hz)
0
1
2
3
4
5
6
7
10^0
10^1
10^2
10^3
10^4
10^5


Step Response
Butterworth
Bessel
Output
Time (seconds)
0
2
4
6
8
10
12
0
0.005
0.01
0.015
0.02
0.025
0.03
0.035
0.04
0.045
0.05


5th Order Butterworth BPF - Magnitude
fc1
fc2
50
0
-50
-100
-150
-200
-250
Magnitude (dB)
10^3
10^4
10^5
10^6
10^7
10^8
Frequency (Hz)


5th Order Butterworth BPF - Phase
fc1
fc2
100
0
-100
-200
-300
-400
-500
-600
-700
-800
-900
Phase (degrees)
10^3
10^4
10^5
10^6
10^7
10^8
Frequency (Hz)

Program Code

```
clc; clear all; close all;

%% HW 8

%% Q1 4th Order Butterworth vs Bessel LPF

% AI NOTICE:
% I had AI help with some of the programming below, particularly group
% delay programming & improving plots to be more legible
% (adding markers for important points, colors & legends)

%% Butterworth Form Per Exam 3 Cheatsheet:
% K / [(s^2+0.765s+1)(s^2+1.848s+1)], plugging in our frequency:

% Kw^4 / [(s^2+0.765ws+w^2)(s^2+1.848ws+w^2)]
% Where w = cutoff frequency, or 2pi * 100 = 628 rad/s and K = 10

% Which gives us:
% 1.55 * 10^12 / [(s^2 + 480.42s + 394384)(s^2 + 1160.5s + 394384)]

N = 4;
fc = 100;
wc = 2*pi*fc;
K = 10;

[BWNumerator, BWDenominator] = butter(N, wc, 's');

BWNumerator = BWNumerator * K;

BWTF = tf(BWNumerator, BWDenominator);

disp(BWTF)

%% Bessel Form


[BesselNumerator, BesselDenominator] = besself(N, wc);

BesselNumerator = BesselNumerator * K;

BesselTF = tf(BesselNumerator, BesselDenominator);

disp(BesselTF)

% Which gives:
% (1.559*10^12) / (s^4 + 1962.8s^3 + 1.7337*10^6s^2 + 7.9403*10^8s +
% 1.5585*10^11)


% DC gain is 10
% 1.5585e12 / 1.5585e11 = 10

% Frequency range
```

```
f = logspace(0,5,2000);
w = 2*pi*f;

% Get frequency response for both filters
[BW_H, ~] = freqs(BWNumerator, BWDenominator, w);
[BE_H, ~] = freqs(BesselNumerator, BesselDenominator, w);

% Find magnitude
BW_Mag = 20*log10(abs(BW_H));
BE_Mag = 20*log10(abs(BE_H));

% Find phase
BW_Phase = unwrap(angle(BW_H))*180/pi;
BE_Phase = unwrap(angle(BE_H))*180/pi;

% Plot magnitude
figure;
semilogx(f, BW_Mag, 'b', f, BE_Mag, 'r', 'LineWidth', 1.5);
grid on;
xlabel('Frequency (Hz)');
ylabel('Magnitude (dB)');
title('Magnitude Response');
legend('Butterworth', 'Bessel');
xlim([1 1e5]);
ylim([-120 40]);

% Plot phase
figure;
semilogx(f, BW_Phase, 'b', f, BE_Phase, 'r', 'LineWidth', 1.5);
grid on;
xlabel('Frequency (Hz)');
ylabel('Phase (degrees)');
title('Phase Response');
legend('Butterworth', 'Bessel');
xlim([1 1e5]);

% Group delay
BW_GroupDelay = -gradient(unwrap(angle(BW_H)), w);
BE_GroupDelay = -gradient(unwrap(angle(BE_H)), w);

% Change to milliseconds
BW_GroupDelay = BW_GroupDelay * 1000;
BE_GroupDelay = BE_GroupDelay * 1000;

% Plot group delay
figure;
semilogx(f, BW_GroupDelay, 'b', ...
         f, BE_GroupDelay, 'r', 'LineWidth', 1.5);
grid on;
xlabel('Frequency (Hz)');
ylabel('Group Delay (ms)');
title('Group Delay');
legend('Butterworth', 'Bessel');
xlim([1 1e5]);
```

```
% Time range for step response
t = linspace(0, 0.05, 2000);

% Get step response
[BW_Step, BW_Time] = step(BWTF, t);
[BE_Step, BE_Time] = step(BesselTF, t);

% Plot step response
figure;
plot(BW_Time, BW_Step, 'b', ...
     BE_Time, BE_Step, 'r', 'LineWidth', 1.5);
grid on;
xlabel('Time (seconds)');
ylabel('Output');
title('Step Response');
legend('Butterworth', 'Bessel');

%% Comparisons:

% Magnitude: Butterworth has a sharp knee for the wc dropoff.
% Phase Response: Butterworth has a slower rise and fall for phase response.
% Group Delay: Butterworth has a sharp rise towards the wc versus a smooth fall.
% Step Response: Butterworth rises and oscillates versus a smooth rise.

%% Q2

% A BW bandpass with:
N = 5;
K = 1;

fc1 = 250e3;
B = 1.5e6; % Bandpass frequencies range (fc1 + B = fc2)

fc2 = fc1 + B; % Allows 250kHz to 1.75MHz

wc1 = fc1*2*pi;
wc2 = fc2*2*pi;

% Center frequency
f0 = sqrt(fc1*fc2); % The squareroot of our cutoffs product is our center frequency.

% Design Butterworth bandpass filter
[BPFNumerator, BPFDenominator] = butter(N, [wc1 wc2], 's');

% Transfer function
BPFTF = tf(BPFNumerator, BPFDenominator);

disp('Butterworth Bandpass Transfer Function:')
disp(BPFTF)

% Frequency range
% Creates 2000 logarithmic values between 10^3 and 10^8
f = logspace(3,8,2000);
w = 2*pi*f;
```

```
% Frequency response
[BPF_H, ~] = freqs(BPFNumerator, BPFDenominator, w);

% Magnitude in dB
BPF_Mag = 20*log10(abs(BPF_H));

% Phase in degrees
BPF_Phase = unwrap(angle(BPF_H))*180/pi;

% Magnitude plot
figure;
semilogx(f, BPF_Mag, 'b', 'LineWidth', 1.5);
grid on;
xlabel('Frequency (Hz)');
ylabel('Magnitude (dB)');
title('5th Order Butterworth BPF - Magnitude');
xline(fc1, '--r', 'fc1');
xline(fc2, '--g', 'fc2');

% Phase plot
figure;
semilogx(f, BPF_Phase, 'b', 'LineWidth', 1.5);
grid on;
xlabel('Frequency (Hz)');
ylabel('Phase (degrees)');
title('5th Order Butterworth BPF - Phase');
xline(fc1, '--r', 'fc1');
xline(fc2, '--g', 'fc2');

% Display cutoff and center frequencies
fprintf('Lower cutoff frequency = %.2f kHz\n', fc1/1000);
fprintf('Upper cutoff frequency = %.2f MHz\n', fc2/1e6);
fprintf('Center frequency = %.2f kHz\n', f0/1000);
```